IMPERIAL

Wildfire Spread Prediction

ML for Tackling Climate Change

Paolo Salvetti
06030218

Ahmed
06047931

Naimeh
06030380

Henry
02229932

Overview

Problem context:

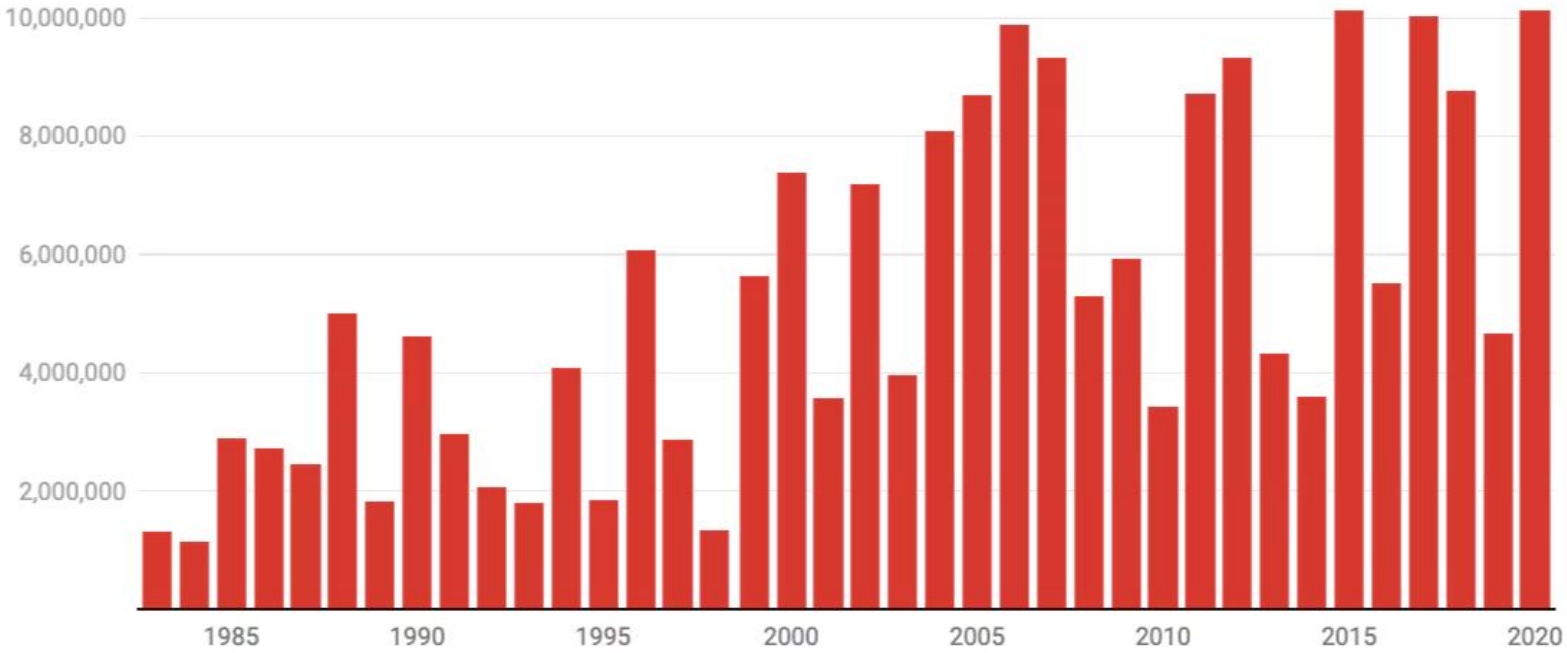
- Wildfires are intensifying globally due to climate change. According to Thomas et al. (NIST, 2017)

- The average annual economic burden from wildfires in the US falls between 71.1 and 347.8 billion USD.

Overview

U.S. wildland area burned each year

More U.S. wildland acres burned in 2015 than any other year on record. The year 2020 was a close second. The majority of these fires are in the West and Alaska.



Prior to 1983, federal wildland fire agencies did not track official wildfire data using current reporting processes.

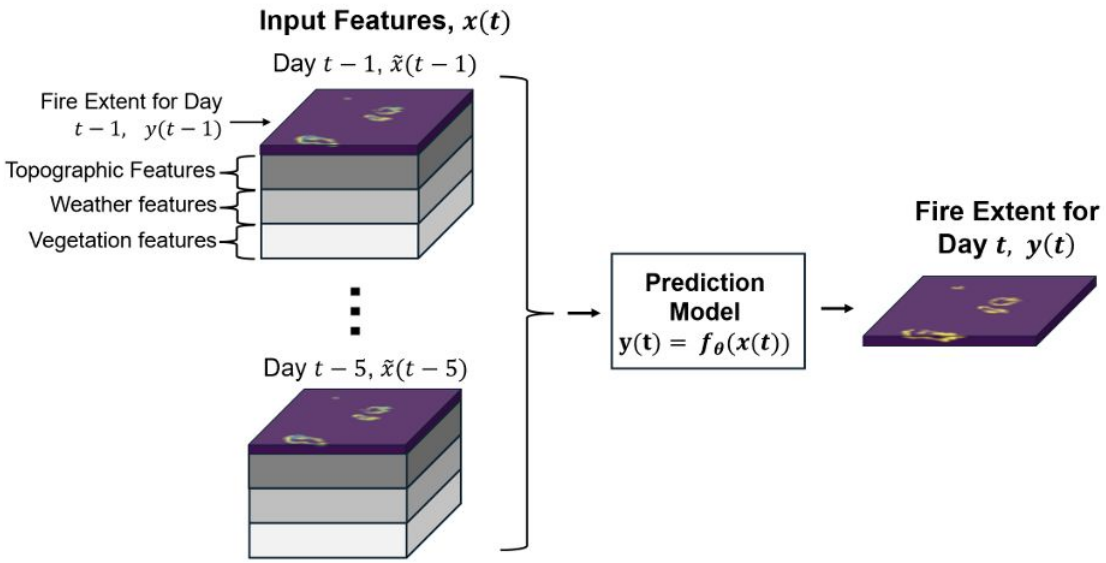
Chart: The Conversation/CC-BY-ND • Source: National Interagency Coordination Center • Get the data

Task

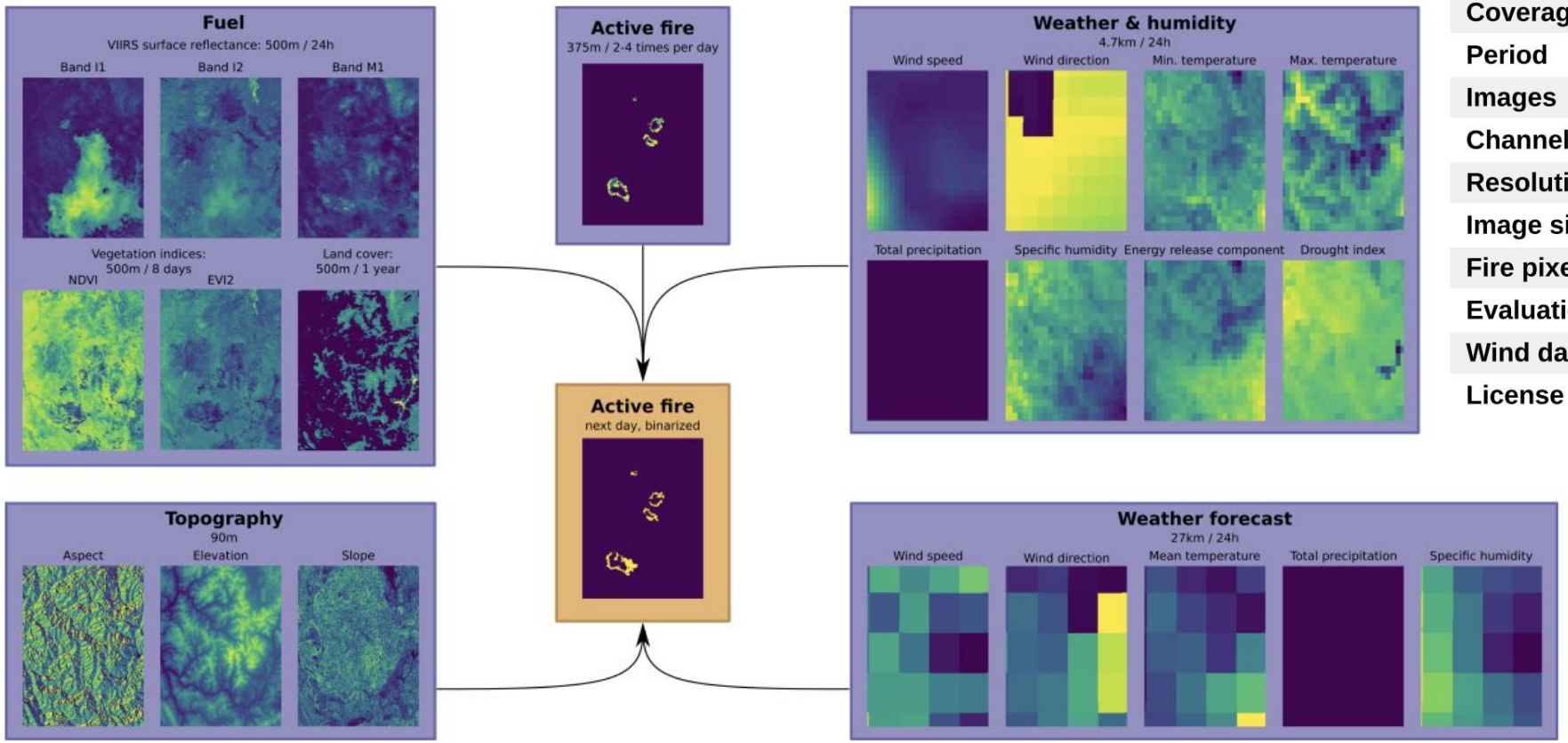
Binary semantic segmentation:

- Given T days of 23-channel geospatial observations (fire, terrain, vegetation, weather) at 375 m resolution, predict a binary mask of active fire on day T+1.

The dataset (Gerard et al., NeurIPS 2023) exhibits extreme class imbalance: only 0.17% of all pixels are active fire. Models are evaluated using Average Precision (AP).



Dataset



Coverage	607 fire events, Western US
Period	January 2018 – October 2021
Images	13,607 daily observations
Channels	23 (fire, terrain, weather, veg.)
Resolution	375 m (VIIRS active fire)
Image size	128 × 128 pixels per sample
Fire pixels	0.17% of total (~1.88M / 1.1B)
Evaluation	12-fold leave-one-year-out CV
Wind data	GRIDMET (obs.) + GFS (forecast)
License	CC BY 4.0, Zenodo

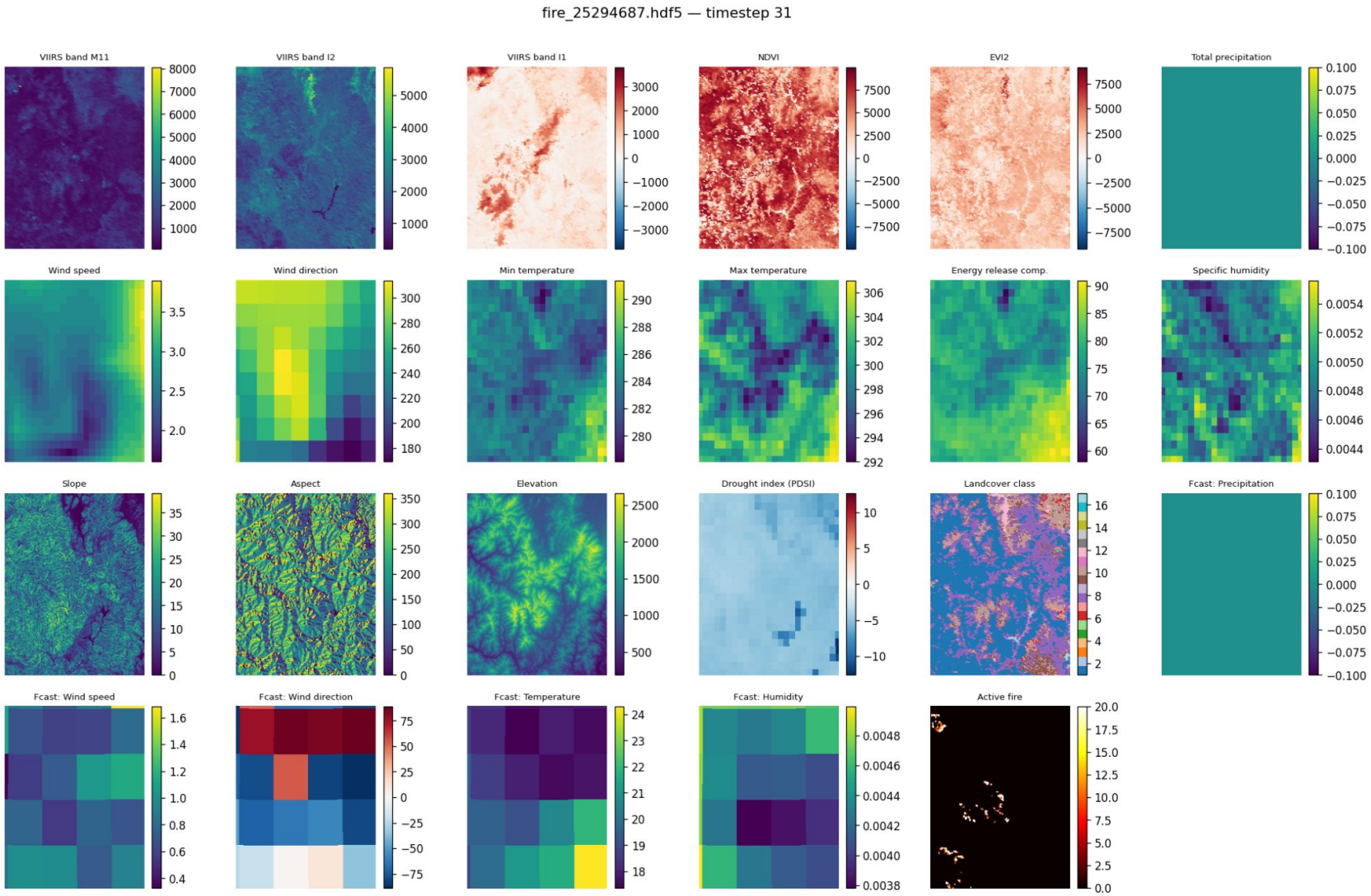
WildfireSpreadTS Dataset

Multi-temporal satellite-derived dataset of wildfires in the United States.

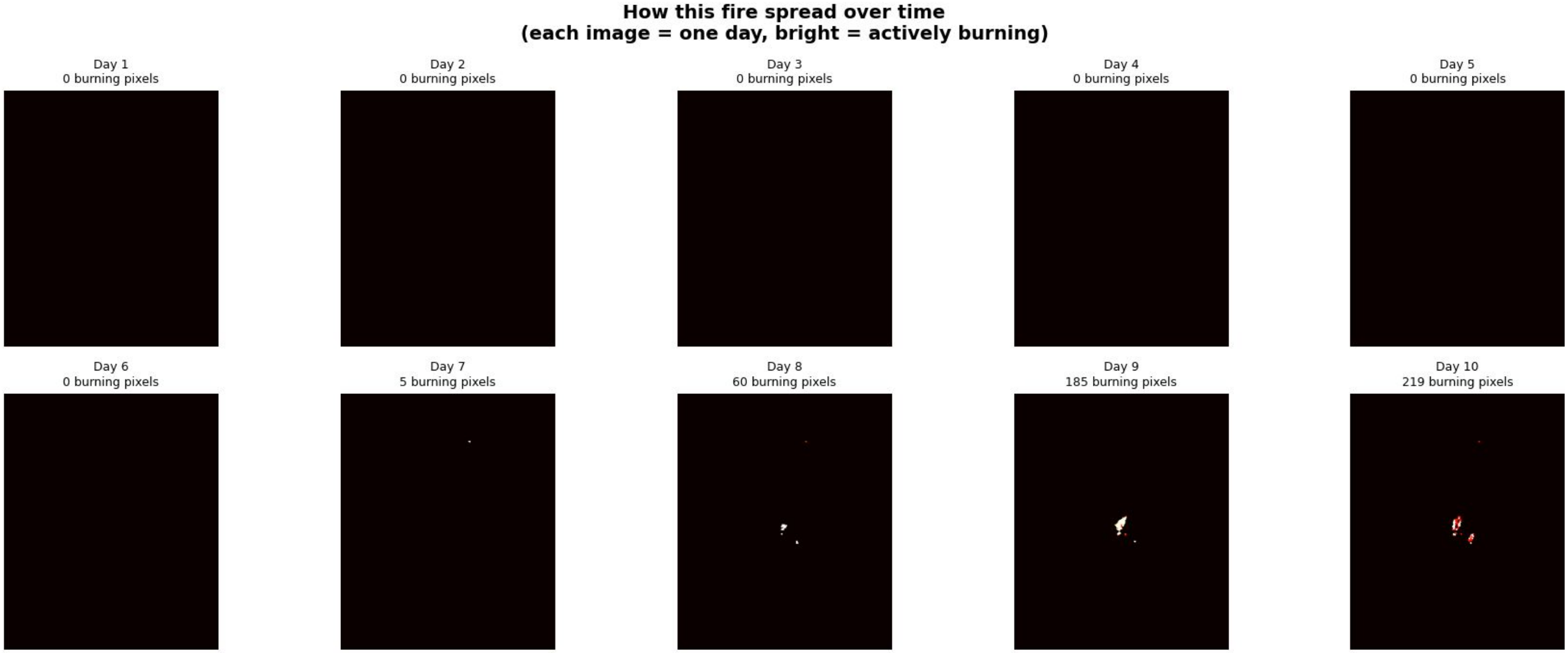
- Each data point is a daily snapshot of an active fire event, cropped to a 128×128 pixel patch (each pixel ≈ 375 m resolution).
- Each HDF5 file stores one fire event that can go for N days and contains the following:
  - x: shape (T, 23, H, W) → 23 geophysical input channels over T consecutive days
  - y: shape (H, W) → binary fire mask for the next day (the prediction target)
    - 23 input channels cover:

Group	Channels	Description
Fire	0	Active fire mask (VIIRS)
Weather (observed	1-7	Total precipitation, wind speed, wind direction (u+v), minimum temperature, maximum temperature, energy release component, specific humidity
Weather (forecast)	8-19	ECMWF next-day forecast for the same variables
Topography	20-21	Elevation, slope
Vegetation	22	NDVI / fuel load proxy

Visualization



Visualization



Experimental Setting

10% of WildfireSpread dataset:

- 3 folds × 20 epochs with batch size = 16
- test metrics: Average precision (handle class imbalance)

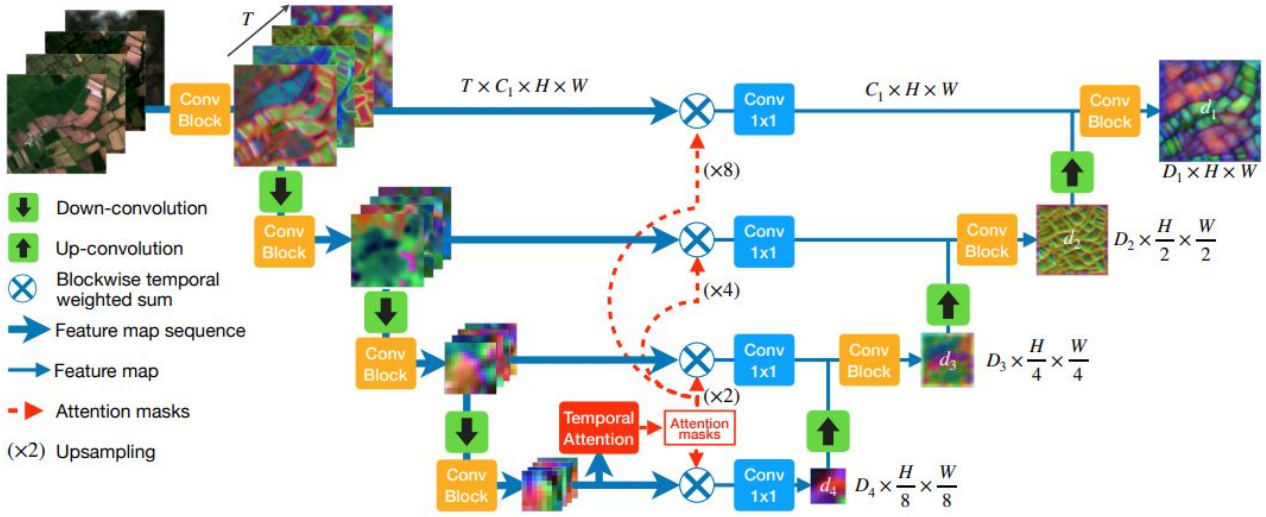
We train & evaluated a variety of model architectures, aiming to provide insights into model designs suitable for this sort of tasks

Fold	train	val	test
0	2018 + 2019	2020	2021
1	2018 + 2020	2019	2021
2	2019 + 2020	2018	2021

SOTA UTAE 2023

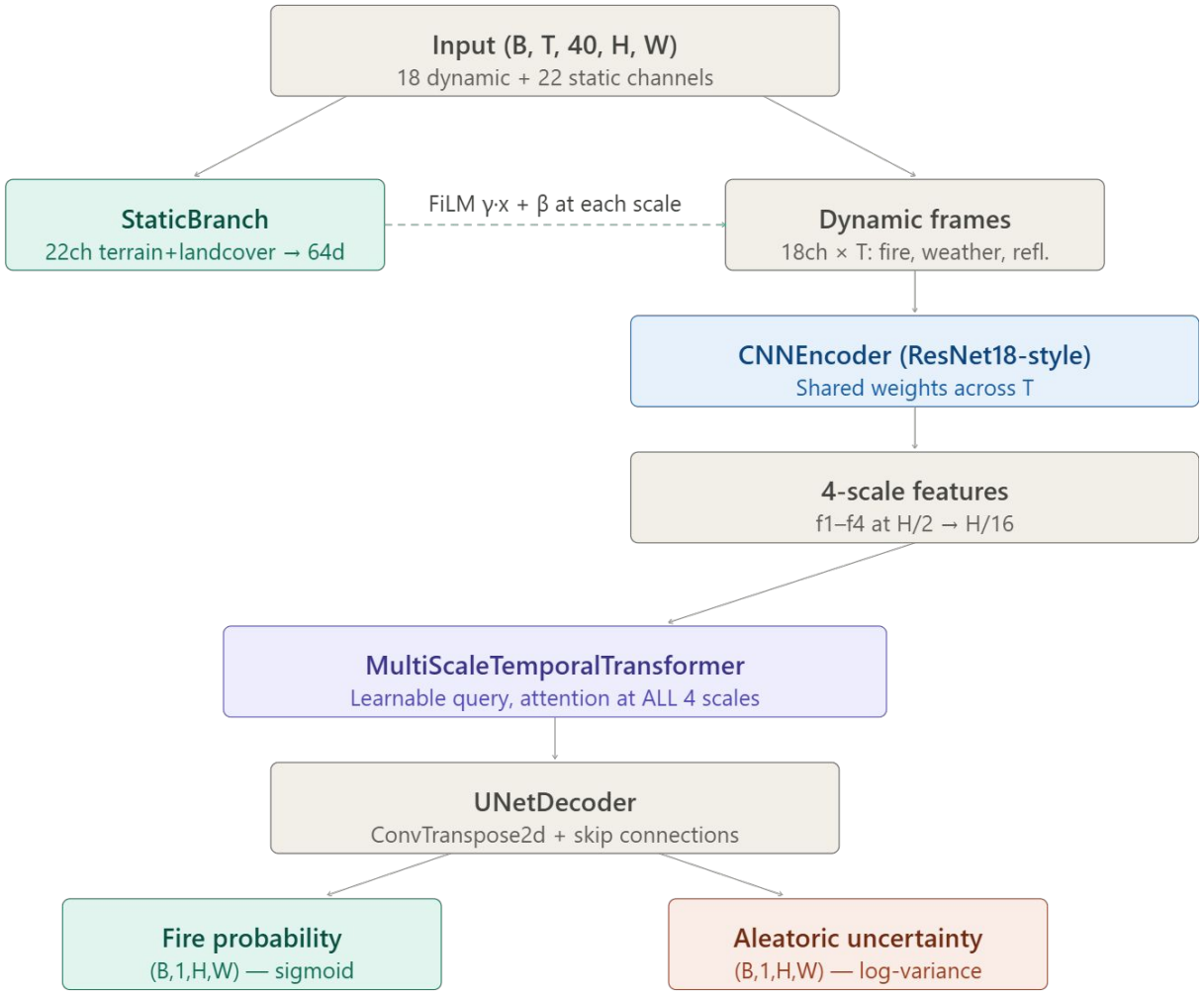
The original WSTS:

- **Res18-UNet**: ResNet-18 encoder (pre-trained on ImageNet) + U-Net decoder with skip connections
- **Input**: 5 consecutive days × 23 channels
- **Model selection** uses Average Precision instead of F1 score on validation set
- Bigger models (ResNet-50, SwinUnet) performed worse, dataset too small, they overfit
- **Key insight**: training recipe matters more than architecture on this benchmark



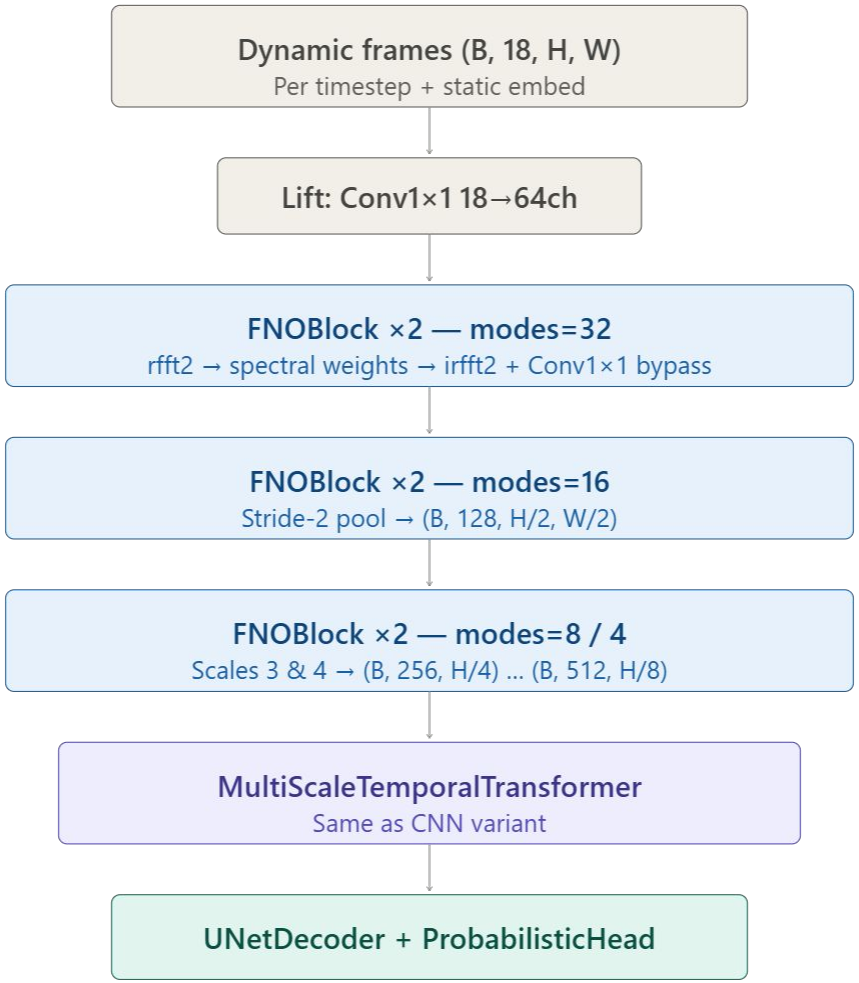
WildfireTransformer — CNN encoder

- Static/dynamic feature separation — terrain & landcover encoded once, injected at every encoder scale via FiLM (γ·x + β). UTAE concatenates everything flat
- Temporal attention at all 4 encoder scales, not just the bottleneck — richer multi-resolution temporal fusion
- ResNet18 encoder (vs UTAE's lightweight custom ConvBlocks) — stronger spatial feature extraction
- Probabilistic output head — two branches: fire probability + aleatoric uncertainty (log-variance) to handle noisy satellite labels
- Focal loss (α=0.97, γ=2) + heteroscedastic loss — better handling of extreme class imbalance (0.1% fire pixels)



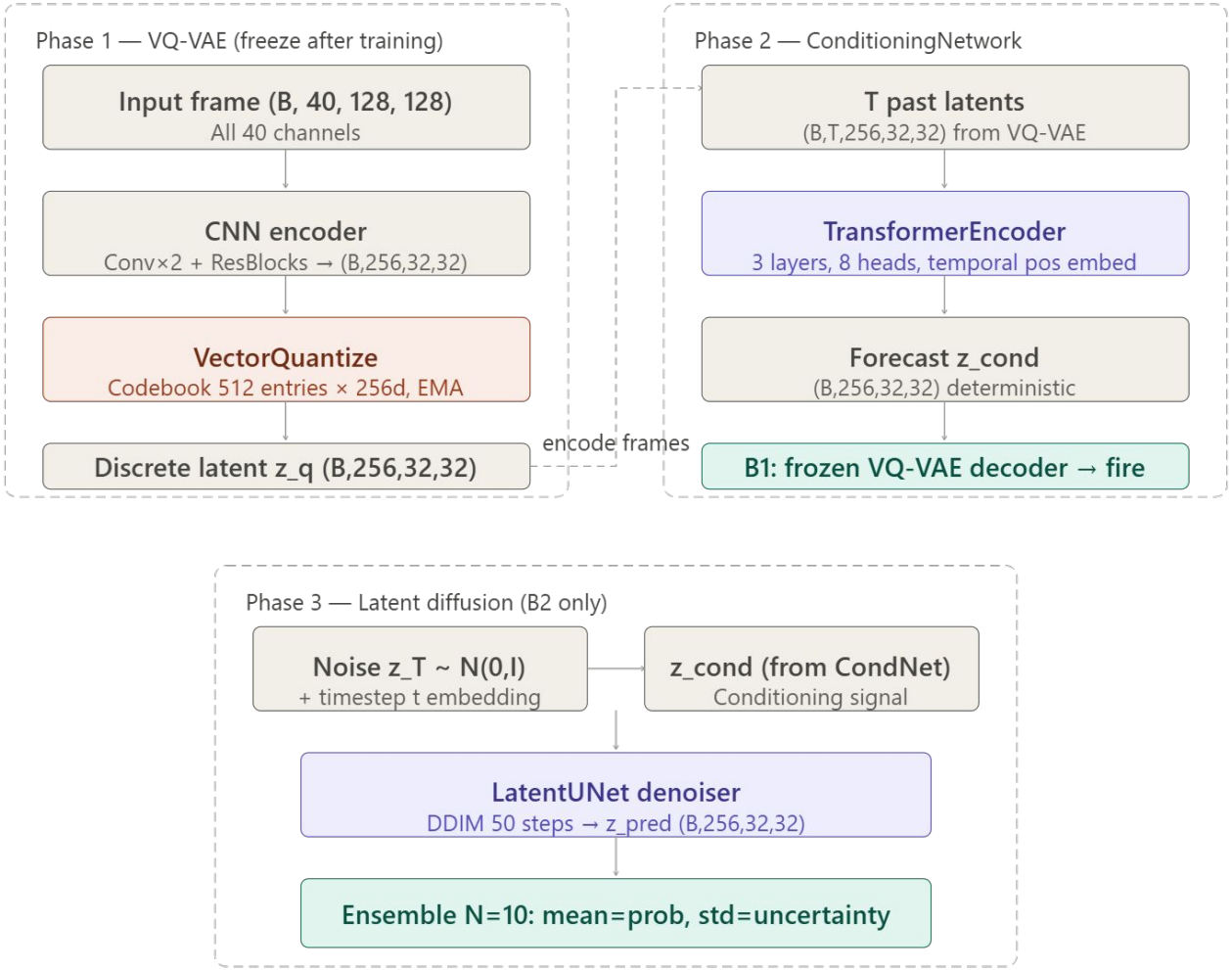
WildfireTransformer — FNO encoder

- Replaces ResNet18 with Fourier Neural Operator — global receptive field from layer 1 via spectral convolution
- Motivated by the 27km weather vs 375m fire resolution mismatch — CNN needs deep stacks to see weather-scale patterns, FNO sees them instantly
- Local Conv1×1 bypass in each FNOBlock preserves sharp fire boundary information

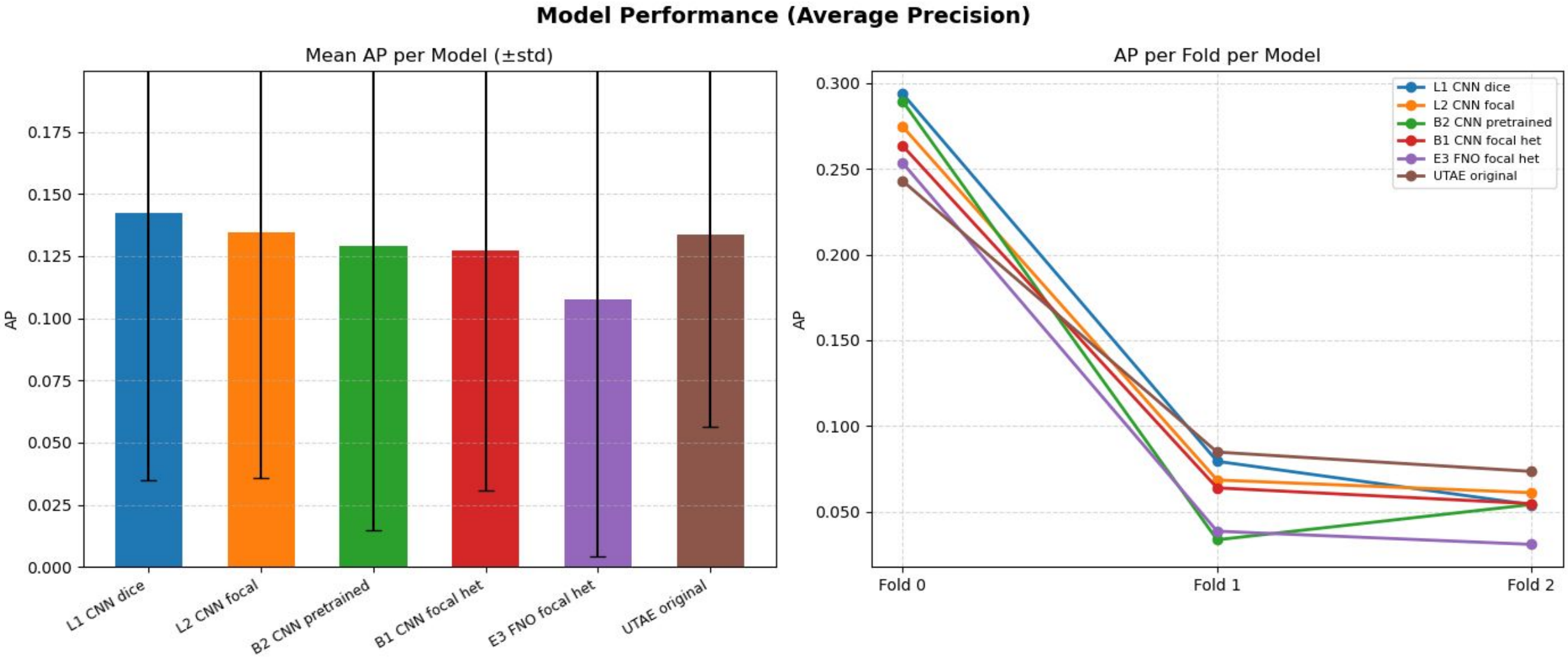


VQ-VAE + ConditioningNetwork

- Generative framing — models the full distribution of plausible fire futures rather than a single deterministic prediction
- Discrete latent space — VQ-VAE compresses 128×128 frames to 32×32 discrete codes; recurring fire/weather patterns become codebook entries
- Temporal modelling in latent space — ConditioningNetwork is a transformer over T latent frames, much cheaper than pixel-space modelling
- B2 adds stochastic diffusion refinement (DDIM, 50 steps) on top of B1's deterministic forecast
- Free uncertainty from ensemble — run inference N=10 times; mean = probability map, std = uncertainty. No separate head needed



Results


Model Performance (Average Precision)
Mean AP per Model (±std)
AP
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
L1 CNN dice
L2 CNN focal
B2 CNN pretrained
B1 CNN focal het
E3 FNO focal het
UTAE original
AP per Fold per Model
AP
0.300
0.250
0.200
0.150
0.100
0.050
Fold 0
Fold 1
Fold 2
L1 CNN dice
L2 CNN focal
B2 CNN pretrained
B1 CNN focal het
E3 FNO focal het
UTAE original

Results

	AP mean	AP std	fold0	fold3	fold10
L1 CNN dice	**0.1424**	0.1076	**0.2939**	0.0793	0.054
L2 CNN focal	0.1347	0.0991	0.2747	0.0684	0.061
B2 CNN pretrained	0.129	0.1141	0.2895	0.0335	0.0541
B1 CNN focal het	0.1273	0.0964	0.2635	0.0638	0.0546
E3 FNO focal het	0.1075	0.1031	0.2535	0.0385	0.0308
UTAE original	0.1337	0.077424	0.243	**0.0847**	**0.0734**
UTAE pretrained CNN	0.098019	0.054053	0.171979	0.077773	0.044305
UTAE pretrained AE encoder	0.031927	0.039348	0.087569	0.004779	0.003435

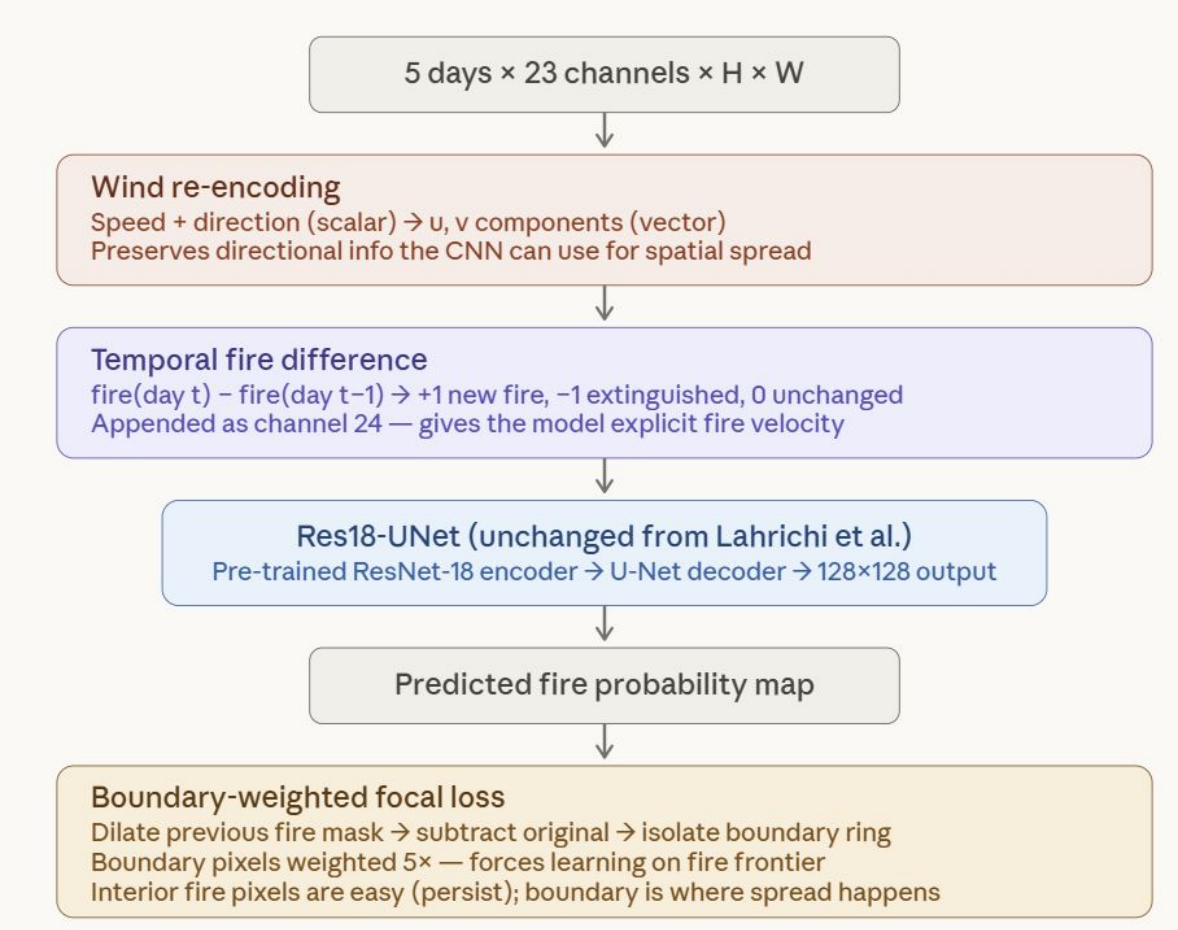
Models - ResNet18 + UNet

No new architecture, no added complexity, just three preprocessing and loss changes:

Wind re-encoding: original data had wind as u/v vectors, dataset authors converted to scalar speed + direction (degrees), we convert back, CNN cannot learn that 270° means "blows east" from a scalar

Temporal fire differences: instead of making the model compare 5 raw fire masks to figure out movement, we compute frame-to-frame differences explicitly — +1 = new fire, −1 = extinguished, 0 = no change.

Boundary-weighted loss: standard focal loss treats all fire pixels equally, but interior pixels are easy (they persist), we upweight the fire boundary ring by 5× where the actual spread prediction happens.



Models - ResNet18 + UNet

Model	Improvements	Input Ch	Fold 0 AP	Fold 1 AP	Fold 2 AP	Mean AP	Δ vs Baseline
Baseline	None	200	0.0095	0.0097	0.0069	**0.0087**	-
Wind u/v only	Wind re-encoding	200	0.0086	0.0113	0.0080	**0.0093**	**+6.9%**
Fire-diff only	Temporal diff	205	0.0124	0.0113	0.0078	**0.0105**	**+20.7%**
Boundary loss only	Boundary weighting	200	0.0102	0.0092	0.0073	**0.0089**	**+2.3%**
Improved (all 3)	All combined	205	0.0071	0.0069	0.0098	**0.0079**	**-9.2%**

Improvement	Hypothesis	Outcome
Wind u/v re-encoding	↑ AP	Weakly confirmed — +6.9% mean but inconsistent across folds; within noise
Temporal fire-diff	↑ AP	**Confirmed** — +20.7% mean, consistent across all 3 folds; statistically the clearest result
Boundary-weighted loss	↑ AP	Not confirmed — +2.3% mean, inconsistent; negligible effect
All three combined	↑↑ AP (synergy)	**Rejected** — improved model is *worse* than baseline (−9.2%); the improvements do not combine synergistically

Models - vit_unet

MiT-B2 (Mix Vision Transformer) encoder + UNet decoder with skip connections

- **Input**: 5 consecutive days × 40 channels, concatenated into 200 channels (data-level fusion) - 40 channels vs paper's 23 due to landcover one-hot encoding and binary fire mask added by the official preprocessing pipeline

- **Loss**: Dice loss with BCE fallback, handles the 0.1% fire pixel imbalance in the 10% subset

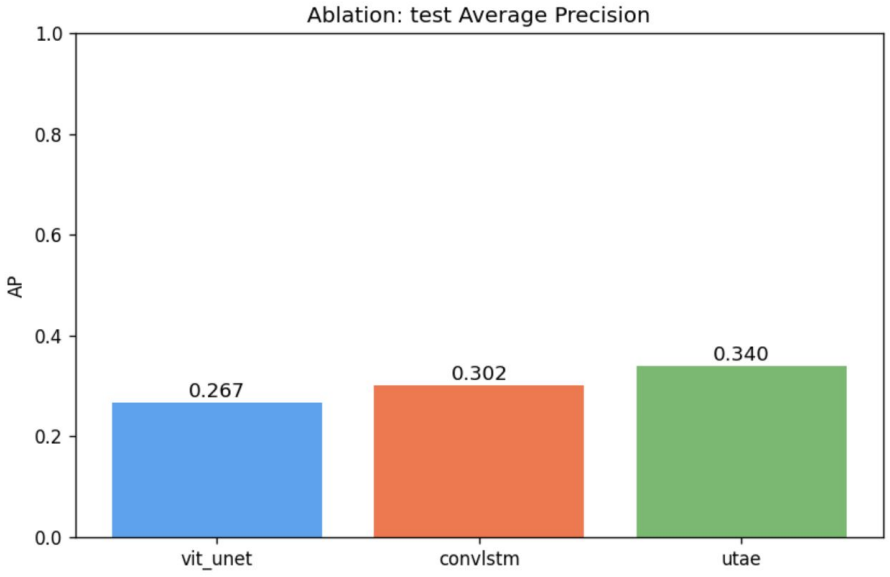
- Checkpoint selection uses val_loss (not val F1), matching the paper's training protocol

- **MiT-B2 vs ResNet-18**: hierarchical ViT produces multi-scale feature maps at 1/4, 1/8, 1/16, 1/32 resolution — richer spatial representations than CNN encoder

- Ablation result (fold 0, seed 42, 20 epochs, bs=128): AP=0.267 - lower than ConvLSTM (0.302) and UTAE (0.340).

UTAE — 3-fold cross-validation results

	test_AP	test_f1	test_iou	test_precision	test_recall	test_loss
0	0.3305	0.0858	0.0448	0.0451	0.8909	0.7171
1	0.1365	0.2599	0.1493	0.1817	0.4558	0.7094
2	0.3987	0.2585	0.1484	0.1529	0.8344	0.7112
mean	0.2886	0.2014	0.1142	0.1266	0.7270	0.7126
std	0.1360	0.1001	0.0601	0.0720	0.2366	0.0040



Models - Temporal Fusion U-Net

Shared encoder: each day processed independently through the same U-Net encoder weights, producing features at 4 levels [64, 128, 256, 512]

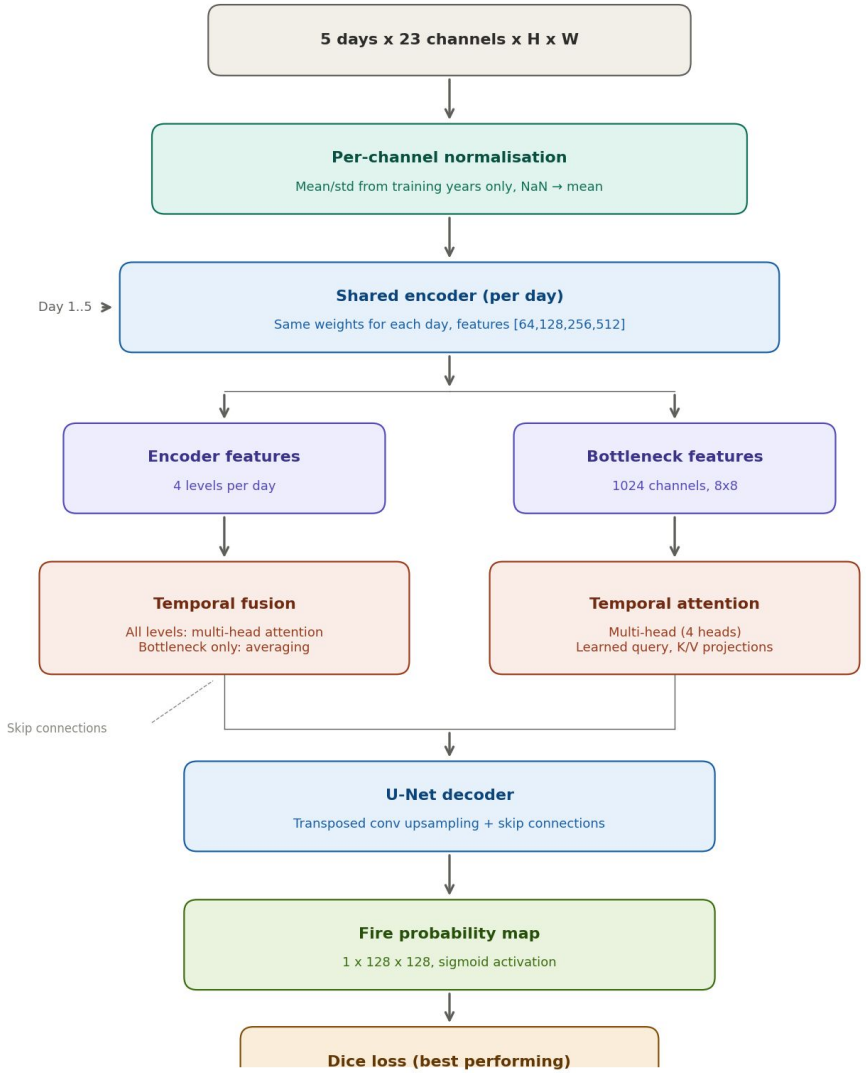
Temporal fusion: multi-head attention (4 heads) fuses features across 5 input days into a single representation before decoding

Two attention variants tested:

- All levels: attention at every encoder level + bottleneck (33.8M params)
- Bottleneck only: attention only at deepest level, averaging elsewhere (33.1M params)

Decoder: standard U-Net decoder with transposed convolutions and skip connections from the fused encoder features

Output: single-channel fire probability map (128 x 128), trained with Dice loss to handle extreme class imbalance (0.1% fire pixels)



Models - Temporal Fusion U-Net

Evaluation Setup

- **Robust Evaluation:** Conducted 3-Fold Cross-Validation across different fire seasons (2019 and 2021) to test generalization.
- **The Baseline (Persistence):** Assumes zero fire spread over 24 hours (T+24h=T0). Acts as a strict "no-learning" reference model.

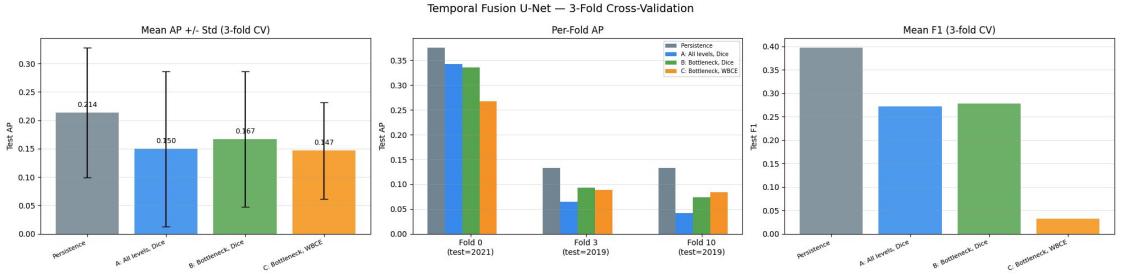
Model Architectures Compared

- **Model A:** Temporal Fusion applied across **all levels** of the U-Net (Dice Loss).
- **Model B:** Temporal Fusion restricted strictly to the **bottleneck** (Dice Loss).
- **Model C:** Bottleneck Temporal Fusion trained with Weighted Binary Cross-Entropy (WBCE) to handle class imbalance.

Key Findings / Takeaways

- **Bottleneck Efficiency:** Fusing temporal data exclusively at the deepest layer (Model B) outperforms multi-level fusion (Model A).
- **High Inter-Fold Variance:** A significant performance drop in 2019 test folds compared to 2021 indicates a severe data distribution shift between years.
- **The Baseline Challenge:** The high Average Precision (AP) of the static Persistence baseline highlights the extreme difficulty of predicting dynamic fire expansion.

Config	Fold 0 test=2021	Fold 3 test=2019	Fold 10 test=2019	Mean ± Std
Persistence baseline	0.376	0.133	0.133	0.214 ± 0.115
A — All levels, Dice	0.342	0.065	0.042	0.150 ± 0.136
B — Bottleneck, Dice	0.336	0.093	0.073	0.167 ± 0.119
C — Bottleneck, WBCE	0.267	0.089	0.084	0.147 ± 0.085



Models - Temporal Fusion U-Net

On the data imbalance problem With ~0.1% fire pixels, any loss function that creates a near-zero gradient at the all-zeros prediction will collapse. The model finds "predict nothing" easier than learning fire patterns.

On regularisation Dropout was the single most impactful intervention, without it, most configs collapsed regardless of loss function, learning rate, or architecture choice.

On model complexity vs data size More parameters and more attention modules hurt rather than help at 10% data scale. Simplicity and stability beat capacity when training samples are scarce.

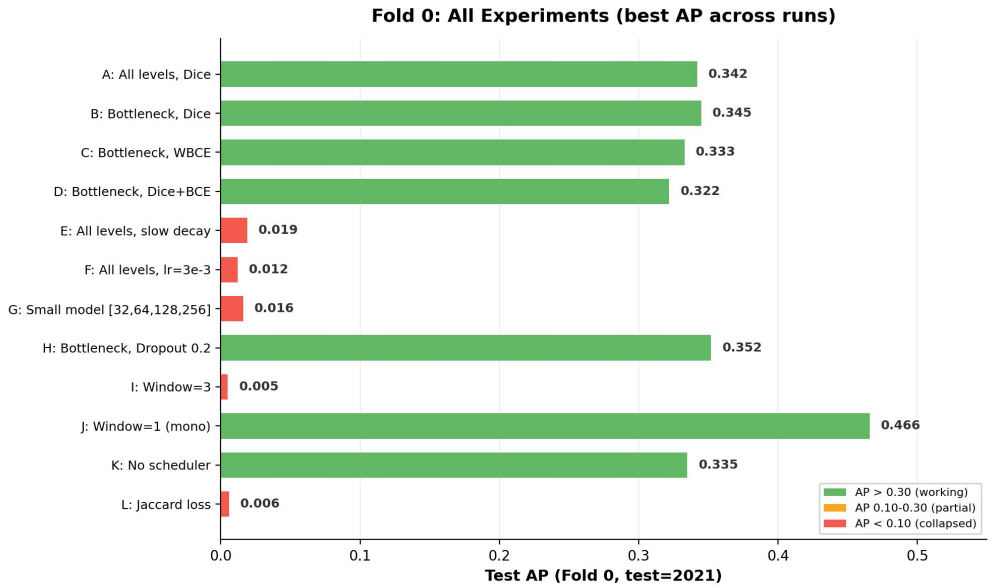
On temporal information More input days does not mean more useful signal, it means more complexity, fewer training samples, and higher collapse risk. On small datasets, temporal fusion can actively harm performance.

On loss function choice Dice, Jaccard, and BCE all fail in different ways on heavily imbalanced data without careful tuning, no loss function is robust out of the box when the positive class is extremely rare.

On reproducibility Without a fixed random seed, results on small datasets are unreliable, two identical runs can produce completely opposite outcomes, making fair comparison between configs very difficult.

On checkpointing Saving the best model by validation loss, not final epoch, is essential, several good results would have been completely lost if training had been evaluated only at the end

Exp	Configuration	Changed	AP	F1	Status
A	All levels, Dice	Attention mode	0.342	0.474	Collapsed ep 27, saved before
B	Bottleneck, Dice	Attention mode	0.345	0.450	Stable
C	Bottleneck, WBCE	Loss: WBCE(500)	0.333	0.082	High recall, low precision
D	Bottleneck, Dice+BCE	Loss: combined	0.322	0.057	High recall, low precision
E	All levels, slow decay	Scheduler: T=100	0.019	0.000	Collapsed ep 15
F	All levels, lr=3e-3	LR: 3e-3	0.012	0.000	Collapsed ep 5
G	Small model	Features: [32,64,128,256]	0.016	0.000	Collapsed ep 14
H	**Bottleneck, Dropout 0.2**	**Dropout: 0.2**	**0.352**	**0.454**	**Best stable model**
I	Window=3	Window: 3	0.005	0.000	Collapsed ep 8
J	**Window=1 (mono)**	**Window: 1**	**0.466**	**0.546**	**Highest AP (unstable)**
K	No scheduler	Scheduler: none	0.335	0.447	Stable
L	Jaccard loss	Loss: Jaccard	0.006	0.000	Collapsed ep 10



Shortcomings

Dataset:

- Too small. Only ~58 fire events across 2018–2020, giving ~350–500 training samples per fold after windowing.
- 10% sample only of WildfireSpreadTS (Rufin et al., 2023).
- Single geography. All fires are in the continental US.
- Temporal and spatial resolutions are coarse.

Model and Training:

- Only 20 epochs
- Fixed threshold
- Class imbalance not fully addressed
- Checkpoint selection

Future Work

Deeper failure mode analysis:

- When and how does our models fail
  - Class imbalance?
  - Learning target off-set?
  - etc.

- Scale up the training dataset
- Explore more architectures
  - Transformer-VQVAEs

Q&A